

Session 07

Getting Started with HTML5

What are you going to learn in this session?

In this session, you will begin your journey into actual web development by learning HTML5 and building your first multi-page website. The focus is not on memorizing tags, but on understanding how a web page is structured and how content is organized.

You will learn how to create an HTML file, understand its basic structure, add text, format content, create lists, and link multiple pages together. By the end of the session, you will have a working personal portfolio website with at least two connected pages.

Why is this topic important?

HTML is the foundation of every website on the internet. No matter how advanced technologies become, every web application ultimately relies on HTML to structure content that the browser can display.

If you do not understand HTML properly, you will struggle with CSS and JavaScript later because they build on top of it. A strong foundation in HTML ensures that you understand how content is organized before you start styling or adding interactivity.

How should you prepare yourself to attend this session?

This is a hands-on session, so you must come prepared to code along with the instructor. Make sure VS Code is installed and working properly, and ensure that your terminal and file system setup from previous sessions are ready.

You should also be comfortable creating folders and files, and navigating your system. Keep your browser open along with VS Code so you can instantly see the output of your work using Live Server.

What should you do after the session?

After the session, you should complete the remaining pages of your portfolio website as practice. This includes adding more sections, improving structure, and linking all pages properly.

You should also revisit all tags covered in class and try to use them in different contexts so that you become comfortable without relying on memorization. The goal is to build familiarity through usage.

Class Notes

What is HTML and How a Web Page Works

HTML stands for HyperText Markup Language, and it is used to define the structure of a web page. It tells the browser what content exists on the page, such as headings, paragraphs, images, and links.

A simple way to understand this is by comparing HTML, CSS, and JavaScript. HTML provides structure, CSS provides styling, and JavaScript provides interactivity. In this session, your focus is only on HTML, so you can clearly understand the foundation.

When you open an HTML file in a browser, the browser reads the file line by line and renders the content visually. This is why correct structure and order are important.

HTML Page Structure

Every HTML page follows a standard structure that helps the browser understand how to interpret the content. You will create this structure quickly using Emmet by typing an exclamation mark and expanding it.

The basic structure includes the declaration, which tells the browser that this is an HTML5 document and ensures proper rendering. It also includes the tag, which wraps the entire page, the section where metadata such as the title is defined, and the section where all visible content is placed.

A simple starter file looks like this:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width,
initial-scale=1.0" />
    <title>My Portfolio</title>
  </head>
  <body>
    <h1>Hello World</h1>
```

```
<p>This is my first HTML page.</p>
</body>
</html>
```

You will build a simple "Hello World" page to understand how these parts work together and how the browser interprets each part of the structure. You will also learn how to write comments, which help you and others understand your code without affecting how it runs.

Here is a simple example of a comment in HTML:

```
<!-- This is a comment. The browser will not display this. -->
```

Building the Class Project (Home Page)

You will start building the home page of your personal portfolio, which will act as your first real project. This page will contain basic information such as your name, introduction, and sections describing your skills or interests.

You will use headings and paragraphs to structure content clearly, along with line breaks and horizontal rules to separate sections visually. You will also learn how to use to group related content, which becomes important as pages grow larger.

A simple portfolio home page may begin like this:

```
<body>
  <div>
    <h1>Sachin Bhatnagar</h1>
    <h2>Aspiring Full Stack Developer</h2>
    <p>Welcome to my personal portfolio website.</p>
    <p>I am learning how to build websites using HTML.</p>
    <hr />
  </div>

  <div>
    <h3>About Me</h3>
    <p>I enjoy technology, coding, and building useful digital
products.</p>
    <br />
    <p>This website is part of my classroom practice.</p>
  </div>
</body>
```

While building, you will use Emmet shortcuts to speed up your workflow and Live Server to instantly view changes in the browser. This helps you connect code with output in real time and makes the learning experience much more practical.

Text Formatting and Lists

HTML provides several tags to format text and make content more meaningful and readable. These include tags for bold text such as `<b>` and `<strong>`, tags for emphasis such as `<i>` and `<em>`, highlighting using `<mark>`, and smaller text using `<small>`.

Here is a simple example showing how these tags may be used inside a page:

```
<p><b>This text is bold.</b></p>
<p><strong>This text is important.</strong></p>
<p><i>This text is in italics.</i></p>
<p><em>This text is emphasized.</em></p>
<p><mark>This text is highlighted.</mark></p>
<p><small>This is smaller text.</small></p>
```

You will also create lists to organize information such as skills or projects. Lists can be unordered using `<ul>` or ordered using `<ol>`, depending on whether sequence matters. While working with these elements, you will also observe the difference between inline and block elements and how they affect layout and spacing.

A simple example of both list types is shown below:

```
<h3>My Skills</h3>
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>

<h3>My Learning Plan</h3>
<ol>
  <li>Learn HTML</li>
  <li>Learn CSS</li>
  <li>Learn JavaScript</li>
</ol>
```

Links and Multiple Pages

Real websites are not limited to a single page, so you will learn how to connect multiple pages using links. This is done using the anchor tag, which allows navigation between pages.

You will create links from the home page to another page such as an About or Projects page, and then build that second page during the session. This helps you understand how multi-page websites are structured and how one page can lead to another.

A simple navigation section may look like this:

```
<a href="index.html">Home</a>
<a href="about.html">About</a>
<a href="projects.html">Projects</a>
```

A basic About page may begin like this:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width,
initial-scale=1.0" />
    <title>About Me</title>
  </head>
  <body>
    <h1>About Me</h1>
    <p>I am learning full stack development and building my first
portfolio.</p>
    <a href="index.html">Go Back to Home</a>
  </body>
</html>
```

You will be encouraged to complete additional pages as practice after the session, reinforcing your understanding through repetition.

Inline Styling (Basic Introduction)

You will get a very light introduction to styling using the style attribute, which allows you to apply simple visual changes directly inside HTML. This is useful for very quick demonstrations, even though it is not the recommended long-term approach.

For example, you can change text color and font size like this:

```
<h1 style="color: blue;">My Portfolio</h1>  
<p style="font-size: 18px;">I am learning HTML step by step.</p>
```

It is important to understand that this is only for demonstration. In real-world development, styling is handled separately using CSS, which will be covered in the next session.

Common Beginner Confusions

Many learners get confused between tags and elements. A tag is the actual syntax like `<h1>`, while an element includes the opening tag, content, and closing tag together.

Another common issue is links not working, which usually happens due to incorrect file paths. You should always check the file name and folder structure carefully.

Live Server issues are also common, so ensure that your file is saved and that the server is running properly before checking the browser.

Key Takeaways

You now understand how to create a basic HTML page and structure content using common tags. You can add headings, paragraphs, lists, and formatted text to make your page meaningful and readable.

You have also learned how to connect multiple pages using links, which is the foundation of building real websites. Additionally, you have been introduced to basic styling and how HTML, CSS, and JavaScript work together at a high level.

At this stage, your goal is not memorization but comfort. The more you build and experiment, the more natural HTML will become.